

Neural Network Complexity Analysis for Data-Driven NARX Identification of a Pneumatic Levitation System

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Abstract—This paper addresses nonlinear system identification using neural networks within a nonlinear autoregressive model with exogenous input (NARX) framework. An unknown discrete-time dynamic system is modeled from 4000 normalized input–output samples acquired at a sampling time of 0.2 s. A systematic study is conducted to evaluate the impact of network architecture complexity on predictive performance by analyzing twelve feedforward multilayer perceptron configurations with one and two hidden layers. Model selection is based on validation and test mean squared error, complemented by one-step-ahead prediction and autonomous free-run simulations. Results show that performance does not improve monotonically with increasing model size; instead, moderate architectures provide the best trade-off between accuracy and generalization. The selected (5,5) architecture achieves the lowest test error while maintaining bounded free-run responses. No overfitting is observed under the adopted dataset size and training strategy, suggesting that moderate network complexity provides favorable generalization behavior for the considered dataset.

Index Terms—System identification, neural networks, NARX model, nonlinear discrete-time systems, multilayer perceptron, model complexity analysis.

I. INTRODUCTION

Data-driven system identification has become an important alternative to first-principles modeling for nonlinear dynamical systems whose internal behavior is difficult to derive analytically. In black-box identification, the objective is to approximate the system input–output mapping directly from experimental measurements without assuming prior structural knowledge. Such approaches are particularly useful in electromechanical systems affected by nonlinearities, unmodeled dynamics, friction effects, and measurement noise. Hybrid analytical and data-driven strategies have demonstrated improved predictive accuracy compared to purely theoretical models in several applications [1], [2].

Among data-driven techniques, artificial neural networks have been widely employed for nonlinear system identification due to their universal approximation capability [3]. In particular, nonlinear autoregressive models with exogenous inputs (NARX) enable feedforward neural networks to represent discrete-time dynamical systems by incorporating past inputs

and outputs as regressors [4]. Owing to their ability to capture complex nonlinear relationships, neural NARX models have become a common approach for black-box dynamic modeling.

Despite their strong approximation capability, increasing neural network complexity introduces a tradeoff between representational capacity and generalization performance. While larger architectures may reduce approximation error, they can also increase variance and susceptibility to overfitting, particularly when finite datasets are available. The implications of this bias–variance tradeoff for neural network design have been extensively discussed in the machine learning literature [5], motivating the need for systematic evaluation of network complexity in nonlinear identification problems.

The experimental dataset used in this work was obtained from a pneumatic levitation system composed of a high-flow fan actuated by pulse-width modulation (PWM) and a lightweight sphere constrained to move vertically inside a transparent tube. The input of the system corresponds to the PWM excitation applied to the fan, while the measured output is the vertical position of the levitated sphere. The dynamics of pneumatic levitation are strongly influenced by nonlinear aerodynamic effects, airflow disturbances, and sensor noise, which complicate the derivation of accurate first-principles models. Consequently, the system constitutes an appropriate benchmark for data-driven black-box identification approaches based solely on measured input–output data.

The objective of this work is to model the nonlinear behavior of the pneumatic levitation system using backpropagation-trained artificial neural networks. The dataset consists of experimentally acquired input–output measurements sampled at 0.2,s, without assuming prior knowledge of the plant structure or governing equations. A nonlinear autoregressive model with exogenous input (NARX-type structure) is implemented using a feedforward multilayer perceptron with five regressors composed of past outputs and present and past inputs. The study evaluates the influence of network complexity by varying the number of neurons in one and two hidden layers while maintaining constant training conditions, including learning rate, activation functions, data partitioning, and optimization

parameters. The central hypothesis is that increasing the number of hidden neurons improves representational capacity and predictive accuracy up to a certain point, beyond which additional complexity may degrade generalization performance. Model selection is performed using the mean squared error (MSE) on a held-out test set, and the identified models are further evaluated through one-step-ahead prediction and autonomous free-run simulation.

The remainder of this paper is organized as follows. Section II presents the theoretical background and methodology, including the problem formulation, neural network architecture, and training procedure implemented in MATLAB. Section III describes the experimental setup, the characteristics of the dataset, and the evaluation metrics employed and discusses the results obtained for training, validation, and testing stages, as well as the effect of network complexity on predictive performance and generalization. Finally, Section IV summarizes the main findings, outlines the limitations of the study, and suggests directions for future research.

II. METHODOLOGY

A. Problem Formulation

Consider an unknown nonlinear discrete-time plant described by the input–output relation

$$y(k) = \mathcal{F}(\varphi(k)) + v(k), \quad (1)$$

where $u(k)$ denotes the input signal, $y(k)$ the measured output, $v(k)$ represents measurement noise, and $\varphi(k)$ is a regressor vector composed of past inputs and outputs. The system is assumed to be causal, nonlinear, and time-invariant. The available dataset consists of 4000 input–output samples collected with a sampling time of $T_s = 0.2$ s.

Prior to model training, both input and output signals are normalized using a Min–Max scaling procedure defined as

$$y_s(k) = \frac{y(k) - y_{\min}}{y_{\max} - y_{\min}}, \quad u_s(k) = \frac{u(k) - u_{\min}}{u_{\max} - u_{\min}}, \quad (2)$$

where $y_{\min}$, $y_{\max}$, $u_{\min}$, and $u_{\max}$ are computed from the dataset. All identification procedures are carried out in the scaled domain.

The specific regressor structure adopted in this work corresponds to a nonlinear autoregressive model with exogenous input (NARX) and is defined as

$$\varphi(k) = \begin{bmatrix} y_s(k-1) \\ y_s(k-2) \\ u_s(k) \\ u_s(k-1) \\ u_s(k-2) \end{bmatrix}. \quad (3)$$

The resulting one-step-ahead predictor is therefore given by

$$\hat{y}_s(k) = f(\varphi(k); \theta), \quad (4)$$

where $f(\cdot)$ is the nonlinear mapping implemented by a multilayer feedforward neural network with parameter vector

θ . During training, a series–parallel configuration is employed, while for multi-step prediction a parallel (free-run) configuration is considered by recursively feeding back predicted outputs.

B. Neural Network Architecture

The nonlinear mapping $f(\varphi(k); \theta)$ is implemented using a fully connected feedforward multilayer perceptron (MLP). The network receives the five-dimensional regressor vector $\varphi(k) \in \mathbb{R}^5$ at its input layer and produces a scalar output $\hat{y}_s(k) \in (0, 1)$ corresponding to the scaled system output.

The architecture consists of one mandatory hidden layer and an optional second hidden layer. Let n_1 and n_2 denote the number of neurons in the first and second hidden layers, respectively, with $n_2 = 0$ indicating the absence of the second hidden layer. All layers are fully connected and include bias terms.

For the general case with two hidden layers, the network equations are given by

$$\mathbf{z}_1(k) = \sigma(\mathbf{W}_1 \varphi(k) + \mathbf{b}_1), \quad (5)$$

$$\mathbf{z}_2(k) = \sigma(\mathbf{W}_2 \mathbf{z}_1(k) + \mathbf{b}_2), \quad (6)$$

$$\hat{y}_s(k) = \sigma(\mathbf{W}_3 \mathbf{z}_2(k) + \mathbf{b}_3), \quad (7)$$

where $\mathbf{W}_1 \in \mathbb{R}^{n_1 \times 5}$, $\mathbf{W}_2 \in \mathbb{R}^{n_2 \times n_1}$, $\mathbf{W}_3 \in \mathbb{R}^{1 \times n_2}$ are the weight matrices, and $\mathbf{b}_1 \in \mathbb{R}^{n_1}$, $\mathbf{b}_2 \in \mathbb{R}^{n_2}$, $\mathbf{b}_3 \in \mathbb{R}$ are the bias vectors.

When the second hidden layer is not present ($n_2 = 0$), the model reduces to

$$\mathbf{z}_1(k) = \sigma(\mathbf{W}_1 \varphi(k) + \mathbf{b}_1), \quad (8)$$

$$\hat{y}_s(k) = \sigma(\mathbf{W}_3 \mathbf{z}_1(k) + \mathbf{b}_3), \quad (9)$$

with $\mathbf{W}_3 \in \mathbb{R}^{1 \times n_1}$.

The activation function used in all hidden and output neurons is the logistic sigmoid function

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (10)$$

which ensures that the network output remains within the interval $(0, 1)$, consistent with the Min–Max normalization applied to the dataset. The logistic sigmoid activation was selected to maintain consistency with the normalized output range and to preserve bounded network outputs during recursive free-run simulation.

The complete parameter vector of the model is therefore defined as

$$\theta = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{W}_3, \mathbf{b}_3\}. \quad (11)$$

Different experimental configurations are obtained by varying n_1 and n_2 , thereby modifying the representational capacity of the neural model while preserving the same regressor structure and training conditions.

The training procedure is summarized in Algorithm 1.

Algorithm 1 Stochastic Backpropagation Training Procedure

```
1: TRAIN( $\mathbf{X}_{\text{train}}, \mathbf{Y}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{Y}_{\text{val}}$ )
2:   Initialize weights  $\mathbf{W}_i$  and biases  $\mathbf{b}_i$  uniformly in
    $[-1, 1]$ 
3:   epoch  $\leftarrow 0$ 
4:   best_val_error  $\leftarrow \infty$ 
5:   while epoch < 2000 and  $\text{MSE}_{\text{train}} > 10^{-4}$  do
6:     for  $k = 1$  to  $N_{\text{train}}$  do
7:       Forward propagate  $\varphi(k)$  through the network
8:       Compute prediction error  $e(k) = y_s(k) - \hat{y}_s(k)$ 
9:       Compute output layer delta using  $\sigma'(x)$ 
10:      Backpropagate deltas to hidden layer(s)
11:      Update all weights and biases using learning rate
       $\alpha = 0.01$ 
12:    end for
13:    Compute  $\text{MSE}_{\text{train}}$ 
14:    Compute  $\text{MSE}_{\text{val}}$ 
15:    if  $\text{MSE}_{\text{val}} < \text{best\_val\_error}$  then
16:      Store current weights as best parameters
17:      best_val_error  $\leftarrow \text{MSE}_{\text{val}}$ 
18:    end if
19:    epoch  $\leftarrow \text{epoch} + 1$ 
20:  end while
21:  return Best stored parameters
```

C. Experimental Design

The experimental evaluation was conducted by systematically varying the number of neurons in the hidden layers while keeping all remaining hyperparameters fixed.

1) *Network Configurations*: The number of neurons in the first hidden layer was selected from

$$N_{\text{ocu1}} = \{2, 3, 5, 11\},$$

while the second hidden layer size was chosen from

$$N_{\text{ocu2}} = \{0, 3, 5\}.$$

A Cartesian product between both sets was performed, yielding 12 architectures.

When $N_{\text{ocu2}} = 0$, the network reduces to a single hidden layer configuration. Otherwise, a two-hidden-layer feedforward structure is used. In all cases, the input dimension is 5 regressors and the output layer contains a single neuron.

2) *Constant Hyperparameters*: All experiments were performed using identical training parameters:

- Learning rate: $\alpha = 0.01$
- Maximum epochs: 2000
- Error tolerance: 10^{-4}
- Activation function: logistic sigmoid (all layers)
- Weight initialization: uniform distribution in $[-1, 1]$
- Bias terms: included in all layers

Input and output signals were scaled using min-max normalization to the interval $[0, 1]$ prior to regressor construction.

Weights were randomly initialized without fixing a random seed. Each configuration was trained once.

3) *Data Partitioning*: A random sample-level partition was adopted to provide a preliminary evaluation of predictive performance across architectures. Although this strategy may introduce temporal dependencies between subsets in dynamic identification problems, it enables consistent comparative analysis under identical training conditions. The partition was done following the conventional 70% training, 15% validation, and 15% testing.

The validation set was used for early stopping by storing the weights corresponding to the minimum validation mean squared error (MSE). Final model performance was computed using the stored best weights.

Although regressors were constructed from sequential past samples, the train/validation/test split was performed randomly over the full dataset.

III. RESULTS AND DISCUSSION

The data set analyzed corresponds to the experimental operation of a pneumatic levitation system in which a sphere is vertically displaced inside a tube through airflow generated by a high-flow PWM-driven fan. The resulting input-output measurements exhibit nonlinear and noisy behavior, providing a suitable benchmark for black-box neural network identification.

All simulations and algorithm implementations were carried out in MATLAB 2024a under identical computational conditions using an Intel i7-based workstation with 16 GB of RAM.

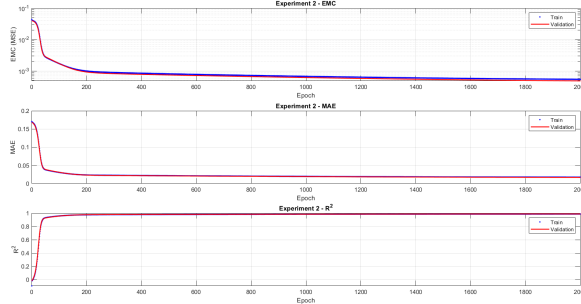
A. Training, Validation, and Test Performance

The learning dynamics were evaluated through the evolution of the mean squared error (MSE), mean absolute error (MAE), and coefficient of determination (R^2) during training. For clarity, two representative architectures are presented: (2, 3) and (5, 5), corresponding to compact and deeper high-performing configurations, respectively.

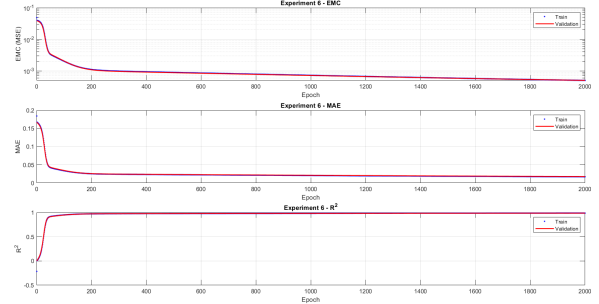
As shown in Fig. 1, both architectures exhibit a monotonic decrease in MSE and MAE throughout training, indicating stable gradient-based optimization under the selected learning rate and initialization strategy. The coefficient of determination increases rapidly during the initial learning phase and progressively approaches unity as prediction error decreases. Training and validation curves evolve similarly across epochs, with no abrupt divergence or instability, suggesting consistent convergence behavior across the evaluated configurations.

All architectures reached the maximum number of epochs (2000), as the stopping tolerance was not triggered. The quantitative performance metrics obtained for the training and validation subsets are summarized in Table I.

The differences between training and validation errors remain small across all configurations, indicating the absence of a significant generalization gap. In several cases, validation errors are slightly lower than training errors, which is consistent with the adopted random partitioning strategy and does not suggest overfitting. The lowest validation MSE was achieved by configuration (2, 3), while the (5, 5) architecture achieved the lowest training MSE.



(a) Architecture (2,3).



(b) Architecture (5,5).

Fig. 1: Training and validation evolution of MSE, MAE, and R^2 for the two representative network architectures.

TABLE I: Training and Validation Performance Metrics for All Network Configurations

Configuration	Train			Validation		
	MSE	MAE	R^2	MSE	MAE	R^2
(2,0)	0.000604	0.018704	0.98493	0.000562	0.018501	0.98563
(2,3)	0.000543	0.017517	0.98647	0.000484	0.016934	0.98763
(2,5)	0.000573	0.018073	0.98572	0.000520	0.017632	0.98670
(3,0)	0.000566	0.018186	0.98589	0.000535	0.018180	0.98632
(3,3)	0.000599	0.018914	0.98506	0.000575	0.018785	0.98531
(3,5)	0.000560	0.017827	0.98603	0.000515	0.017297	0.98686
(5,0)	0.000571	0.018246	0.98576	0.000541	0.018159	0.98618
(5,3)	0.000556	0.017772	0.98614	0.000502	0.017193	0.98717
(5,5)	0.000498	0.017134	0.98758	0.000500	0.017767	0.98724
(11,0)	0.000567	0.018212	0.98586	0.000537	0.018198	0.98628
(11,3)	0.000594	0.018290	0.98518	0.000543	0.017832	0.98613
(11,5)	0.000641	0.019507	0.98401	0.000615	0.019658	0.98429

To further evaluate predictive capability, all models were assessed using the held-out test subset, corresponding to 15% of the available dataset. The resulting performance metrics are presented in Table II.

TABLE II: Test Set Performance for All Architectures

Architecture	MSE	MAE	R^2
(2,0)	0.00060147	0.019156	0.98482
(2,3)	0.00054728	0.018003	0.98619
(2,5)	0.00056966	0.018551	0.98562
(3,0)	0.00055695	0.018536	0.98594
(3,3)	0.00058903	0.019140	0.98514
(3,5)	0.00057097	0.018429	0.98559
(5,0)	0.00056803	0.018659	0.98567
(5,3)	0.00055886	0.018300	0.98590
(5,5)	0.00049079	0.017448	0.98761
(11,0)	0.00056691	0.018682	0.98569
(11,3)	0.00059910	0.018910	0.98488
(11,5)	0.00063051	0.019738	0.98409

Among the evaluated models, the (5,5) configuration achieved the lowest test MSE and MAE, as well as the highest R^2 , indicating the best overall predictive performance on unseen data. Although the (2,3) architecture obtained the best

validation MSE during model selection, its test performance remained comparable to that of (5,5), suggesting that both architectures provide similar predictive performance.

Increasing network complexity did not produce a monotonic improvement in performance. In particular, configurations with 11 neurons in the first hidden layer did not outperform smaller architectures and, in some cases, exhibited slightly degraded validation and test metrics. Similarly, the inclusion of a second hidden layer did not consistently improve predictive accuracy across all neuron combinations. These results suggest that moderate network complexity provides sufficient representational capacity for the considered identification problem while maintaining stable generalization behavior.

B. Qualitative Prediction Behavior

To qualitatively assess the modeling accuracy of the selected model, Fig. 2 presents the one-step-ahead output estimation obtained using the (5,5) architecture over 100% of the available dataset. The figure shows the measured output (solid black line), the network estimation (dashed red line), and the system input (blue dotted line).

A strong visual alignment between the measured and estimated outputs can be observed across the entire time horizon. The predicted signal closely overlaps the real output, indicat-

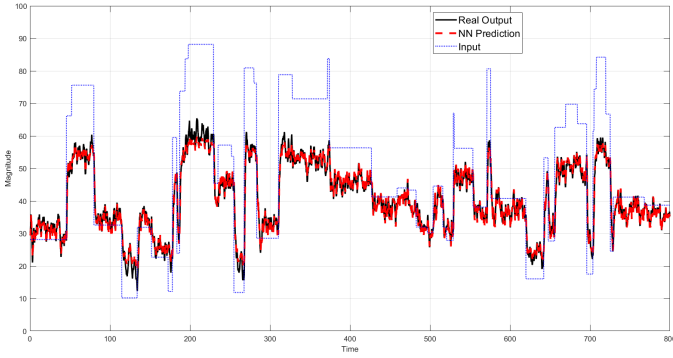


Fig. 2: One-step-ahead prediction using the (5,5) architecture over the full dataset.

ing that the network successfully captures the dominant system dynamics under one-step-ahead operation.

The model accurately follows transient regions characterized by fast rising and falling edges. No noticeable temporal lag is observed between the estimated and measured signals, suggesting adequate dynamic responsiveness. However, during abrupt amplitude variations, the network exhibits slight smoothing behavior. In particular, sharp peaks are mildly underestimated and appear slightly flattened compared to the measured output. Importantly, no overshoot behavior is detected.

The output signal is visibly noisy, and the network does not replicate high-frequency noise components entirely. Instead, the estimation presents a smoother profile, indicating that the model primarily captures the underlying system dynamics rather than fitting measurement noise. Despite this smoothing effect, the predicted signal remains centered correctly, with no observable systematic bias or offset.

C. Free-Run Simulation

1) *Free-Run Setup*: To evaluate the autonomous estimation accuracy of the selected (5,5) architecture, a free-run simulation was performed. In this mode, the network operates recursively: the predicted output at each time step is fed back as part of the regressor vector for the next prediction. Specifically, the regressor is defined as $[y(k-1) \ y(k-2) \ u(k) \ u(k-1) \ u(k-2)]^T$, where past outputs correspond exclusively to previously estimated values.

Unlike the one-step-ahead configuration, no measured output is used after initialization. The model is therefore fully autonomous. The external input consists of a piecewise-constant excitation sequence (20, 55, and 15 units, each applied for 30 samples), scaled according to the training normalization parameters. Initial conditions for both input and output states were set to zero, representing a rest condition.

2) *Free-Run Results and Stability Analysis*: Figure 3 presents the free-run response of the network under the described excitation profile.

The predicted output remains bounded and stable throughout the entire simulation horizon. No divergence, oscillatory

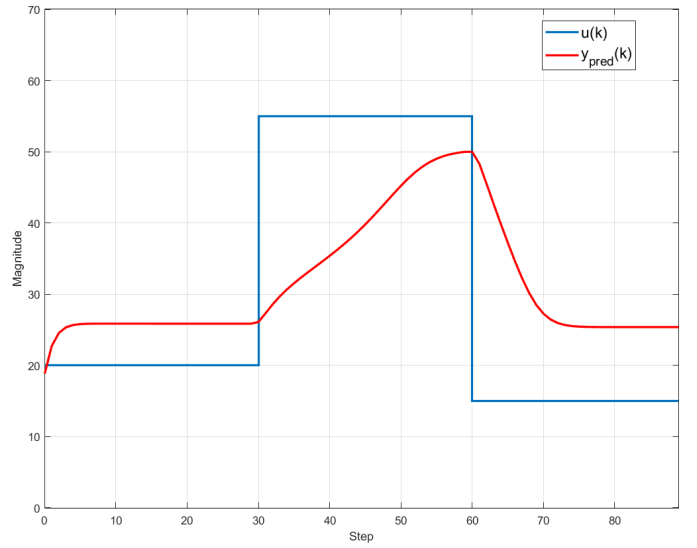


Fig. 3: Free-run simulation using the (5,5) architecture under piecewise-constant input excitation.

growth, or saturation behavior is observed. The network responds consistently to each step change in the external input, preserving the qualitative nonlinear relationship between input and output magnitude.

The dynamic response appears coherent across all three excitation levels. The output transitions smoothly between operating regions and stabilizes after each step change without exhibiting instability or artificial oscillations. This indicates that the recursive structure does not introduce numerical drift or amplification of internal states.

It is important to note that this experiment does not aim to replicate previously measured trajectories. Instead, it evaluates the internal dynamic consistency of the learned model when operating autonomously under a new input sequence. In contrast to the one-step-ahead scenario, where measured outputs constrain the prediction at each step, the free-run mode tests whether accumulated prediction errors destabilize the system. The observed bounded behavior suggests that the learned nonlinear mapping preserves coherent recursive dynamics over the evaluated simulation horizon.

IV. CONCLUSIONS

This work presented a data-driven nonlinear system identification approach based on feedforward multilayer perceptrons with a NARX-type regressor structure. The influence of network complexity was evaluated by comparing multiple one- and two-hidden-layer configurations under identical training conditions using experimental data acquired from a pneumatic levitation system.

The obtained results show that increasing network size does not necessarily improve predictive performance. Among the evaluated configurations, the (5,5) architecture achieved the best overall test performance while maintaining stable behavior in both one-step-ahead prediction and free-run simulation. In contrast, larger architectures did not consistently provide

improved generalization, indicating that moderate network complexity was sufficient to capture the dominant nonlinear dynamics of the considered system.

The similarity between training, validation, and test errors across configurations suggests stable generalization behavior under the adopted training strategy. Additionally, the free-run simulations demonstrated bounded autonomous predictions under recursive operation, indicating that the identified model preserves coherent dynamic behavior beyond one-step-ahead estimation.

Although the results are promising, several limitations remain. The dataset was randomly partitioned despite the temporal nature of the problem, and each architecture was trained only once without statistical analysis of initialization variability. Future work should therefore consider structured time-series validation strategies, repeated training experiments, alternative activation functions, and regularization techniques.

Comparisons against classical nonlinear identification approaches and validation under additional excitation conditions may further improve the robustness and generalization assessment of the proposed methodology.

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